

Memory & Memory Organisation

Address & data lines · capacity · RAM, SRAM, DRAM · ROM family · hierarchy & cache · memory chip design

QUICK TRICK

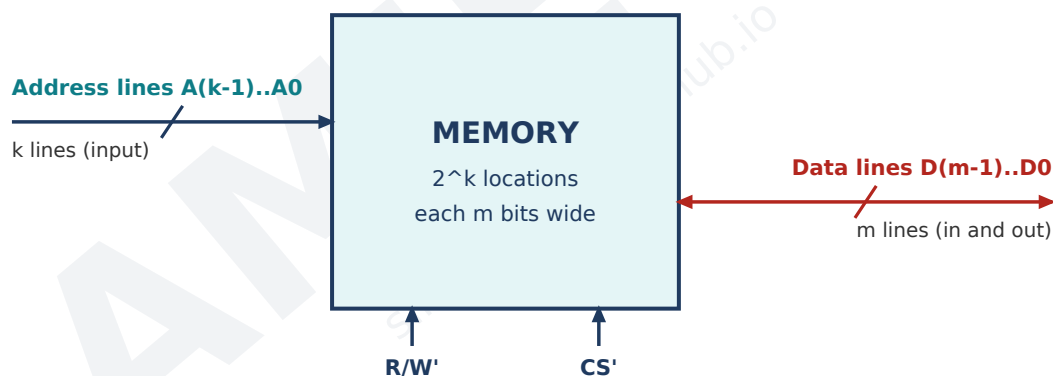
Three lines to learn first: **Locations** = $2^{(\text{address lines})}$. **Capacity** = **locations** × **data width**. **Chips needed** = **(total size) / (chip size)**.

16.1 Memory basics

● LEVEL 1 · BEGINNER

What is memory? Memory is where a computer stores information (programs and data). It is a big collection of storage cells. Each **location** holds one **word** (a group of bits) and has its own **address** (a number).

Analogy. A building with numbered rooms. The **room number** is the **address**. What is inside the room is the **data**. To read, you give a room number and look inside. To write, you give a room number and put something inside.



Line	What it does	How many?
Address lines	Choose which location to use (input)	k lines for 2^k locations
Data lines	Carry the data in and out	m lines for an m -bit word
R/W' (read/write)	1 = read, 0 = write	1
CS' (chip select)	Switches the chip on (usually active low)	1 or more

EASY MEMORY

Address lines choose WHERE. Data lines carry WHAT. Address lines are inputs only. Data lines work both ways.

16.2 Memory size, address lines and data lines

● LEVEL 2 • EXAM

Unit	Value
1 K (kilo)	$2^{10} = 1024$
1 M (mega)	$2^{20} = 1,048,576$
1 G (giga)	$2^{30} = 1,073,741,824$
1 byte	8 bits

Number of locations = 2^k (k = number of address lines)
Capacity (in bits) = locations x word width
Address lines k = $\log_2$ (locations)

QUICK TRICK

Splitting trick for address lines: write the size as a power of 2 using K, M, G. **32 K = $2^5 \times 2^{10} = 2^{15}$ → 15 lines.** 4K → 12 lines. 64K → 16 lines. 1M → 20 lines. 16M → 24 lines. 4G → 32 lines. (K adds 10, M adds 20, G adds 30. The number in front gives its own power: 2 → 1, 4 → 2, 8 → 3, 16 → 4, 32 → 5, 64 → 6.)

Example 1 (Level 1): a 4K × 8 memory

What does 4K mean? $4 \times 1024 = 4096$ locations.

Address lines: $4096 = 2^{12} \rightarrow 12$ lines. **Data lines: 8** (each word has 8 bits).

Total bits: $4096 \times 8 = 32,768$ bits = 4096 bytes = **4 KB**.

Example 2 (Level 2): a 64K × 16 memory

Address lines: $64K = 2^6 \times 2^{10} = 2^{16} \rightarrow 16$. Data lines: **16**.

Total bits: $2^{16} \times 16 = 2^{20} = 1$ Mbit = 2^{17} bytes = **128 KB**.

Example 3 (Level 3): byte-addressable memory

A computer has 32-bit words and a **byte-addressable** memory of 4 GB. Each byte has its own address.

4 GB = $2^2 \times 2^{30} = 2^{32}$ bytes → **32 address lines**.

Number of 32-bit words = $2^{32} / 4 = 2^{30}$.

Address ranges in hexadecimal

Size	Number of locations	Last address if it starts at 0000
1 K	400 (hex)	03FF
2 K	800	07FF
4 K	1000	0FFF
8 K	2000	1FFF
64 K	10000	FFFF

QUICK TRICK

Last address = first address + size – 1 (size in hex). Example: a 1K block starting at 4000 ends at $4000 + 3FF = 43FF$. Remember: $1K = 400_{16}$, $2K = 800_{16}$, $4K = 1000_{16}$.

16.3 RAM: SRAM and DRAM

● LEVEL 1 • BEGINNER

RAM (Random Access Memory) can be **read and written**, and any location can be reached in the same time. RAM is **volatile**: the data is **lost when the power is off**. There are two types.

	SRAM (Static RAM)	DRAM (Dynamic RAM)
Memory cell	A flip-flop, about 6 transistors	1 transistor + 1 capacitor
How it stores a bit	The flip-flop keeps its value while powered	Charge in a capacitor (leaks away)
Refresh needed?	No	Yes (every few milliseconds)
Speed	Faster	Slower
Size of a cell / density	Bigger cell, fewer bits per chip	Smaller cell, many bits per chip
Cost per bit	Higher	Lower
Power	More (per bit)	Less when idle, but refresh uses power
Used for	Cache memory , registers	Main memory (the RAM in a computer)
Address pins	All address bits at once	Multiplexed : row address, then column address

QUICK TRICK

S = Stable (no refresh), D = Dynamic (refresh). SRAM: fast, costly, small → cache. DRAM: slow, cheap, big → main memory. **Cell count**: 6 transistors for SRAM, 1 transistor + 1 capacitor for DRAM.

Why does DRAM need refresh? The capacitor slowly loses its charge, so the bit would fade away. A refresh circuit **reads and rewrites** every row regularly (for example, every 64 ms). Reading a DRAM cell also destroys the charge, so it must be rewritten after each read.

GATE POINT

DRAM address multiplexing: a DRAM sends the **row address** first (with RAS') and then the **column address** (with CAS') on the **same pins**. So the number of address pins is **half**. Example: a 16M × 1 DRAM needs 24 address bits but only **12 pins**.

16.4 ROM family

● LEVEL 1 • BEGINNER

ROM (Read Only Memory) is **non-volatile**: it **keeps its data without power**. It is used for fixed programs, like the start-up program (BIOS) of a computer.

Type	Full name	How is it programmed?	How is it erased?
ROM	Read Only Memory	At the factory (mask)	Cannot be erased
PROM	Programmable ROM	By the user, once	Cannot be erased
EPROM	Erasable Programmable ROM	By the user	Ultraviolet (UV) light (the whole chip, through a window)
EEPROM	Electrically Erasable PROM	Electrically	Electrically , one byte at a time
Flash	Flash memory (a type of EEPROM)	Electrically	Electrically, in blocks (fast and cheap)

QUICK TRICK

Letters tell the story: **P** = Programmable (once). **EPROM** = **E**rasable with **UV**. **EEPROM** = **E**lectrically **E**rasable. **Flash** = EEPROM that erases in blocks, so it is dense and cheap (USB drives, SSDs).

RAM	ROM
Read and write	Mostly read (EPROM/EEPROM/Flash can be rewritten, slowly)
Volatile	Non-volatile
Working storage (programs and data in use)	Permanent storage (start-up code, firmware)
Fast writing	Slow or limited writing

COMMON MISTAKE

Saying EPROM is erased electrically. EPROM uses **UV light**. EEPROM uses **electricity**.

16.5 Memory chip design (expansion)

● LEVEL 2 • EXAM

You often need a memory bigger than the chips you have. You can join small chips in two ways, or both:

Expansion	Goal	How
Width (word size)	More bits per word	Put chips side by side . They share the same address lines. Each gives different data lines.
Depth (number of locations)	More locations	Stack chips. Use the extra address lines and a decoder to select one chip (CS').

QUICK TRICK

Chips needed = (required total size) / (chip size). More exactly: **rows** (depth) = required locations / chip locations, **columns** (width) = required word width / chip word width. Chips = rows × columns.
Extra address lines for the decoder = $\log_2(\text{rows})$. These are the **top** address lines. The chips themselves use the lower lines.

Example 1 (Level 2): 4K × 8 from 1K × 4 chips

Rows = 4K / 1K = 4. Columns = 8 / 4 = 2. **Chips** = 4 × 2 = 8.

Address lines = 12. Chips use 10 (A9–A0). The decoder uses A11, A10 (2-to-4).

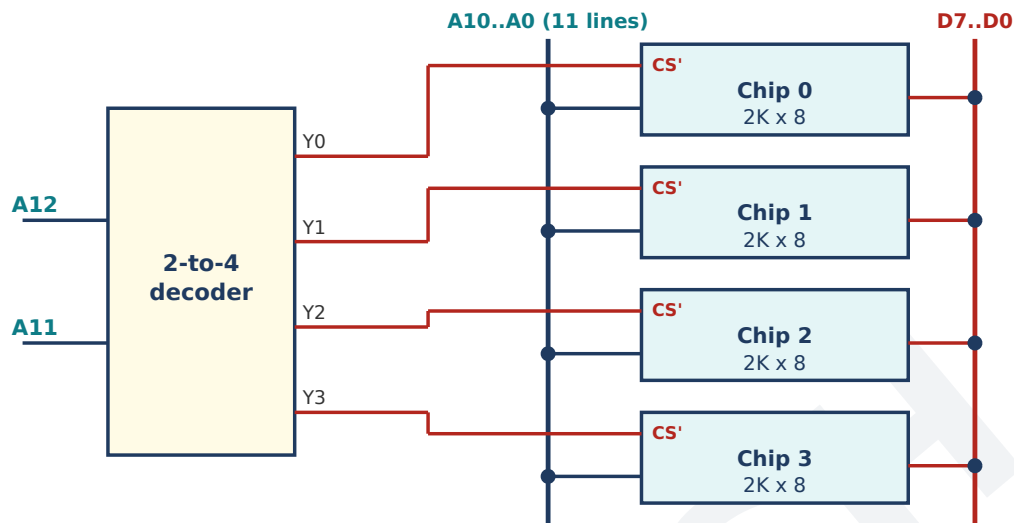
Example 2 (Level 3): 16K × 16 from 4K × 8 chips

Rows = 16K / 4K = 4. Columns = 16 / 8 = 2. **Chips** = 8.

Address lines = 14 (16K = 2^{14}). Each chip needs 12 (A11–A0). The extra lines A13, A12 go to a **2-to-4 decoder** that selects 1 of 4 rows. In each row, one chip gives D15–D8 and the other gives D7–D0.

Worked design: 8K × 8 from four 2K × 8 chips

Total address lines: $8K = 2^{13} \rightarrow 13$ (A12–A0). Each chip has $2K = 2^{11}$ locations $\rightarrow$ A10–A0. The extra lines A12 and A11 go into a 2-to-4 decoder. Each decoder output selects one chip. All chips share the address lines A10–A0 and the data lines.



A12 A11	Chip selected	Address range (hex)
0 0	Chip 0	0000 – 07FF
0 1	Chip 1	0800 – 0FFF
1 0	Chip 2	1000 – 17FF
1 1	Chip 3	1800 – 1FFF

QUICK TRICK

Reading the memory map: each chip covers $2K = 0800_{16}$ addresses, so the chips start at 0000, 0800, 1000, 1800. The chip number is the value of the top (decoder) address bits.

GATE POINT

Tri-state outputs: all chips share one data bus. A chip whose CS' is not active turns its outputs **off** (high impedance, Z). Only the selected chip drives the bus.

Chip select logic for one chip (address decoding)

Example (Level 3): where does a 1K chip sit?

A system has a 16-bit address (A15–A0). A $1K \times 8$ chip uses A9–A0. We want it at **4000 – 43FF**.

$4000_{16} = 0100\ 0000\ 0000\ 0000$. The 6 upper bits A15–A10 = **0 1 0 0 0 0** for every address in 4000–43FF.

CS = $A15' \cdot A14 \cdot A13' \cdot A12' \cdot A11' \cdot A10'$ (a 6-input AND with NOT gates on the 0 bits).

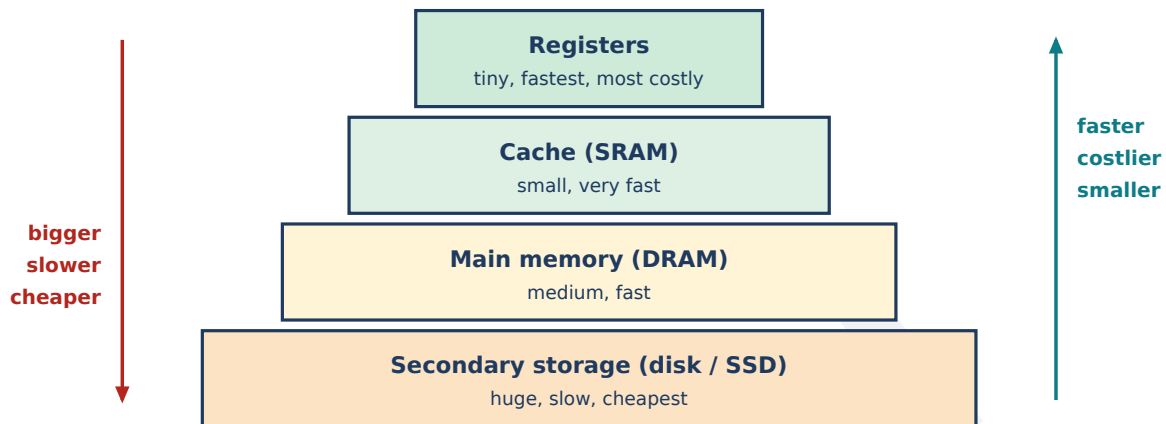
QUICK TRICK

Full decoding uses **all** upper address lines, so each address matches only one location. **Partial decoding** ignores some lines (cheaper), so the same chip appears at several addresses (**mirror images**) and wastes address space.

16.6 Memory hierarchy and cache

● LEVEL 2 • EXAM

Problem. Fast memory is expensive and small. Cheap memory is big but slow. **Idea.** Use a **hierarchy**: a little fast memory close to the processor and lots of slow memory further away. The computer works as if it had a big, fast memory.



Cache is a small, fast SRAM between the processor and main memory. It keeps copies of the data most likely to be used again. This works because of **locality of reference**: **temporal** (a used item will be used again soon) and **spatial** (items near a used item will be used soon).

Hit : the data is in the cache.

Miss : the data is not in the cache (go to main memory).

Average access time = $h \times t_{\text{cache}} + (1 - h) \times t_{\text{main}}$ (h = hit ratio)

Example (Level 3)

Given: hit ratio $h = 0.9$, cache time = 10 ns, main memory time = 100 ns.

Concept: weighted average.

Calculation: $0.9 \times 10 + 0.1 \times 100 = 9 + 10 = 19$ ns.

Answer: 19 ns (much better than 100 ns).

Quick-Solve Tricks — Summary

Topic	Trick
Address lines	Write the size as $2^a \times K/M/G$. $K = 10$, $M = 20$, $G = 30$. Add the powers.
Capacity	Locations $\times$ data width (bits). Divide by 8 for bytes.
Address range	Last = first + size – 1. $1K = 400h$, $2K = 800h$, $4K = 1000h$.
SRAM vs DRAM	SRAM: 6T, fast, no refresh, cache. DRAM: 1T + 1C, refresh, main memory.
DRAM pins	Address pins are half (row and column multiplexed).
ROM family	PROM once. EPROM UV. EEPROM electrical (byte). Flash electrical (block).
Number of chips	Rows $\times$ columns = (total size) / (chip size).
Decoder inputs	$\log_2(\text{rows})$ top address lines.
Chip select	AND of the upper address bits (complement the 0 bits).

Topic	Trick
Average access time	$h \times t_{\text{cache}} + (1 - h) \times t_{\text{main}}$.

GATE QUICK CHECK — Chapter 16

Locations = 2^k Capacity = $2^k \times m$ bits $K=2^{10}$, $M=2^{20}$, $G=2^{30}$
 Chips = (required size) / (chip size) Decoder lines = $\log_2(\text{rows})$
 Average access time = $h t_c + (1 - h) t_m$

Traps: bits vs bytes · byte-addressable vs word-addressable · EPROM (UV) vs EEPROM (electrical) · DRAM address pins are multiplexed · the decoder uses the TOP address lines.

Practice questions

● LEVEL 1 · BEGINNER

- Q1.** How many address lines are needed for 1K locations?
Q2. What does “4K × 8” mean? Find the total number of bits.
Q3. Which memory needs to be refreshed: SRAM or DRAM?

● LEVEL 2 · EXAM

- Q4.** A memory is 64K × 16. Find the address lines, the data lines and the capacity in KB.
Q5. How many 1K × 4 chips are needed to make a 4K × 8 memory?
Q6. An 8 KB memory starts at address 0000. What is its last address (hex)?

● LEVEL 3 · GATE

- Q7.** A 16K × 16 memory is built from 4K × 8 chips. How many chips? What decoder is needed and which address lines does it use?
Q8. A 1K × 8 chip must occupy addresses 4000–43FF in a 16-bit address space. Write its chip-select expression.
Q9. A 16M × 1 DRAM has multiplexed addresses. How many address pins does it have?
Q10. A cache has hit ratio 0.95, cache access time 5 ns and main memory access time 100 ns. Find the average access time.
Q11. A byte-addressable memory of 4 GB uses 32-bit words. How many address lines does it have, and how many words?
Q12. Which ROM type is erased by ultraviolet light, and which can be erased one byte at a time electrically?

Answers

Q	Solution
1	$1K = 2^{10} \rightarrow 10$ lines.
2	$4K = 4096$ locations, each 8 bits. Total = $4096 \times 8 = 32,768$ bits (4 KB).
3	DRAM (the capacitor leaks).
4	Address: $64K = 2^{16} \rightarrow 16$ lines. Data: 16 lines. Capacity: $2^{16} \times 16$ bits = 2^{20} bits = 2^{17} bytes = 128 KB .
5	Rows = $4K/1K = 4$, columns = $8/4 = 2 \rightarrow 8$ chips.
6	$8K = 2000_{16}$ locations. Last = $0000 + 2000 - 1 = 1FFF$.

Q	Solution
7	Rows = $16K/4K = 4$, columns = $16/8 = 2 \rightarrow$ 8 chips . Extra lines = $\log_2 4 = 2 \rightarrow$ a 2-to-4 decoder on A13 and A12 . (Chips use A11-A0.)
8	The chip uses A9-A0. For 4000-43FF the upper bits A15-A10 = 010000. CS = A15' A14 A13' A12' A11' A10' .
9	$16M = 2^{24} \rightarrow$ 24 address bits. Multiplexed (row + column) $\rightarrow$ 12 pins .
10	$0.95 \times 5 + 0.05 \times 100 = 4.75 + 5 =$ 9.75 ns .
11	4 GB = 2^{32} bytes $\rightarrow$ 32 address lines. Words = $2^{32} / 4 = 2^{30}$ words.
12	EPROM (UV light). EEPROM (electrical, byte by byte). Flash erases electrically in blocks.

Next: Chapter 17 — PLA, PAL and PLD (programmable logic).

AMJESH
singhamjesh.github.io